CHRISTIAN CORONA

Robotics & Machine Learning

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OBJECTIVE:

Systems driven machine learning engineer eager to specialize in robotic control, hardware integration, sim2real, and perceptive autonomy. Passionate about developing and deploying robust, multi skill RL policies and learned representation pipelines for platform agnostic robotic systems including quadrupeds, manipulation platforms, drones, and bipeds.

SKILLS:

- Robotics & Simulation: Isaac Lab / Mujoco, ROS2, Perceptive Locomotion, Modular Systems.
- Machine Learning & Control: Deep Reinforcement Learning, Policy Distillation, Spatiotemporal Sequence Modeling, supervised learning, filtering, PyTorch, JAX, ONNX.
- Programming: C++, C, Python, C#, NVIDIA Jetson Deployment

EXPERIENCE:

IHMC Robotics Lab | AI Engineer Intern - AI Controllers Team | Pensacola, FL | Summer 2026
Advisor: Dr. Geoffrey Clark

- C++/ONNX Deployment: Engineered low latency C++ deployment pipelines to execute ONNX exported Deep RL policies directly on NVIDIA Jetson compute modules, transporting motor commands and state feedback over high frequency Ethernet packets.

- Multi Skill RL Policy Design: Designed end-to-end Reinforcement Learning controllers in Isaac Lab for quadrupedal platforms, specifically Unitree Go2, creating unified multi skill policies capable of executing running and jumping without discrete mode switching.

    - Formulated command structures inspired by (arXiv:2602.00678), establishing nominal running commands and modeling jumping as dynamic offsets driven by a step signal (-1) decaying across takeoff, flight, and post-flight phase dependent rewards.
    - Structured reward function weights and curriculum learning stages to stabilize multi terrain agility and dynamic velocity tracking for proprioceptive policies.

- LiDAR Perceptive Locomotion Architecture: Developed a LiDAR based perceptive locomotion framework for terrain awareness.

    - Adapted the encoder to decoder approach and teacher student distillation architecture for 3D point cloud LiDAR inputs to build short term terrain memory and ensure consistency under perceptual degradation. The encoder to decoder approach was trained to reconstruct the ground truth height map while the control policy was trained in simulation.

- Cross Platform Dynamics: Focused primary work and hardware deployment on the Unitree Go2 quadruped while acquiring operational knowledge of the architectures of IHMC's in house bipedal robot.

Ongoing Research | University of Florida (**NSF HAIC** Project) | Gainesville, FL | Fall 2026 - Present

- Safe RL Exoskeleton Control & Sim2Real: Developing safe reinforcement learning control policies for sim2real transfer on exoskeleton platforms as part of the NSF HAIC project, collaborating with Control Barrier Function safety researchers.

- Musculoskeletal Simulation & Neuromechanics: Building high fidelity biomechanical simulation environments using MyoSuite musculoskeletal human models in mjlab / MuJoCo, and exploring imitation learning to mimic human neuromechanical reactions to external perturbations and robotic assistance.

EDUCATION:

Master of Science in Electrical and Computer Engineering
University of Florida, Gainesville, FL
August 2026 - May 2028

- Fall 2026: Digital signal processing, Machine Learning

Bachelor of Science in Computer Science, *Magna cum laude*
Belhaven University, Jackson, MS
April 2026

- Relevant coursework: Deep Learning, Differential Equations, Number Theory, Graph Theory.

PROJECTS:

Physics-Informed Neural Networks (PINN) | *Undergraduate Capstone* | GitHub

- Built a JAX based PINN framework enforcing physical conservation laws to solve complex ODE systems (6 compartment dynamic SEIRD models) with sparse training data under Dr. Jarrod Mau (Belhaven University).

ATHLETICS & LEADERSHIP:

College Baseball Pitcher (Belhaven University / Freed-Hardeman University)

- Competed in high level collegiate athletics and the Appalachian MLB sponsored summer league, developing strong performance under pressure, teamwork, and time management skills while maintaining high academic honors.